

Functions and Graphs

A function is always dependent on one or more variables, like time, distance or a more abstract quantity.

Functions of the form $y = ax + q$

Functions with a general form of $y = ax + q$ are called straight line functions. In the equation, $y = ax + q$, **a** and **q** are constants and have different effects on the graph of the function. The general shape of the graph of functions of this form is shown in Figure below for the function $f(x) = 2x + 3$

To find x and y

Step 1

For y -intercept let $x = 0$

Substitute $x = 0$ where there is x in the equation $y = 2x + 3$

$$y = 2(0) + 3$$

$$y = 3$$

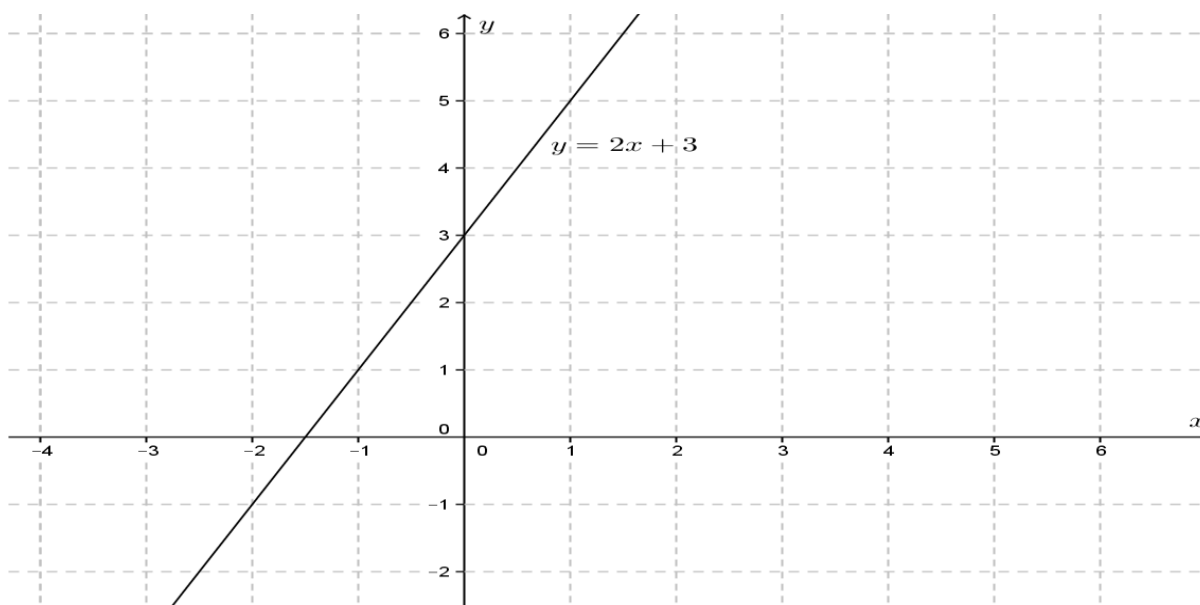
For x -intercept let $y = 0$

Substitute $y = 0$ where there is y in the equation $y = 2x + 3$

$$0 = 2x + 3$$

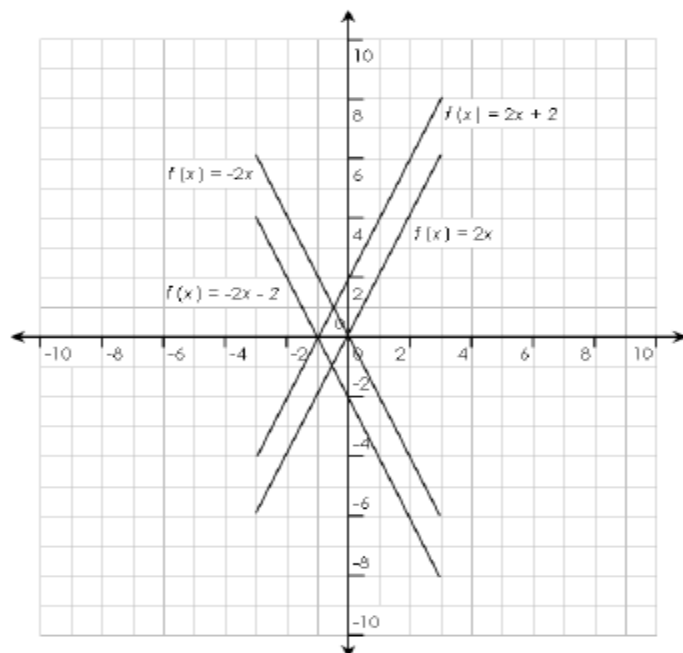
$$-2x = 3$$

$$-x = \frac{3}{2}$$



You can sketch the above graph using Table Method

1. $f(x) = 2x$, $f(x) = -2x$, $y = 2x + 2$ and $y = -2x - 2$



x	-3	-2	-1	0	1	2	3
$2x$	-6	-4	-2	0	2	4	6
$-2x$	6	4	2	0	-2	-4	-6
$2x + 2$	-4	-2	0	2	4	6	8
$-2x - 2$	4	2	0	-2	-4	-6	-8

Parabola

Functions of the Form:

$$y = ax^2 + q$$

The general shape and position of the graph of the function of the form $f(x) = ax^2 + q$, called a parabola, is shown in Figure 11.8. These are *parabolic* functions.

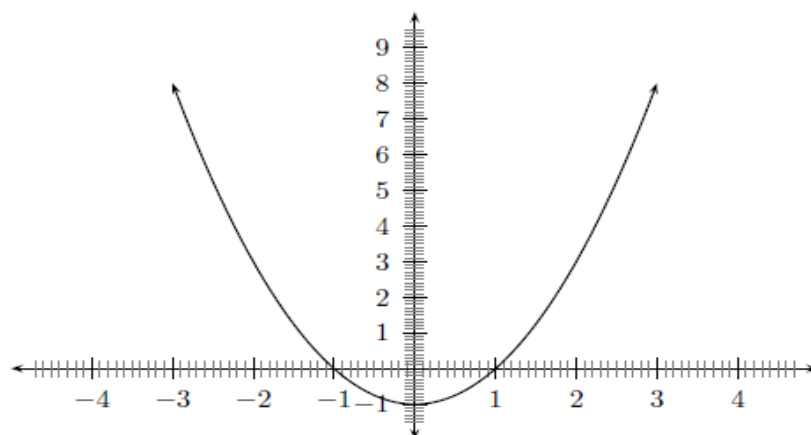


Figure 11.8: Graph of $f(x) = x^2 - 1$.



$a > 0$ (a positive smile)

$a < 0$ (a negative frown)

Figure 11.9: Distinctive shape of graphs of a parabola if $a > 0$ and $a < 0$.

$$y = x^2$$

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

$$f(-3) = (-3)^2 = 9$$

$$f(-2) = (-2)^2 = 4$$

$$f(-1) = (-1)^2 = 1$$

$$f(0) = (0)^2 = 0$$

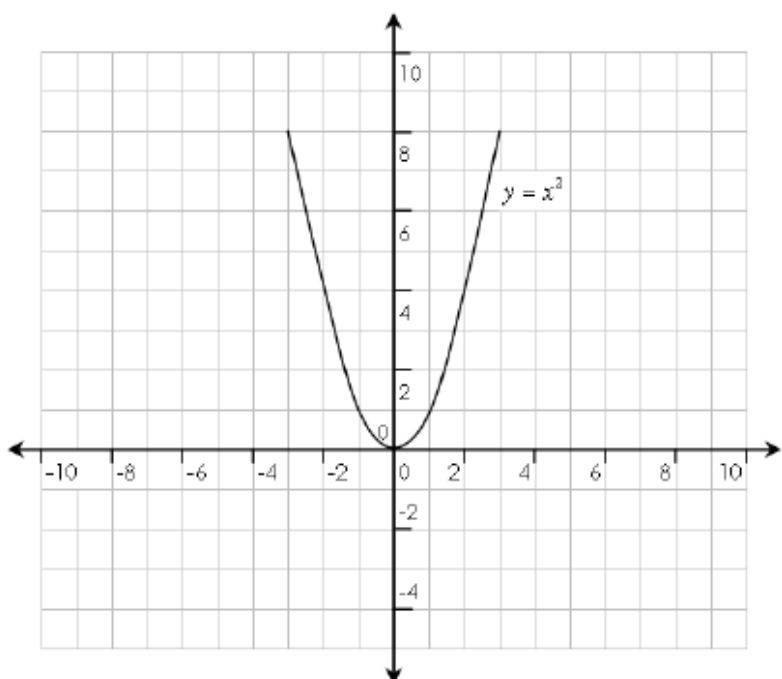
$$f(1) = (1)^2 = 1$$

$$f(2) = (2)^2 = 4$$

$$f(3) = (3)^2 = 9$$

Domain: $\{x : -3 \leq x \leq 3\}$

Range: $\{y : 0 \leq y \leq 9\}$

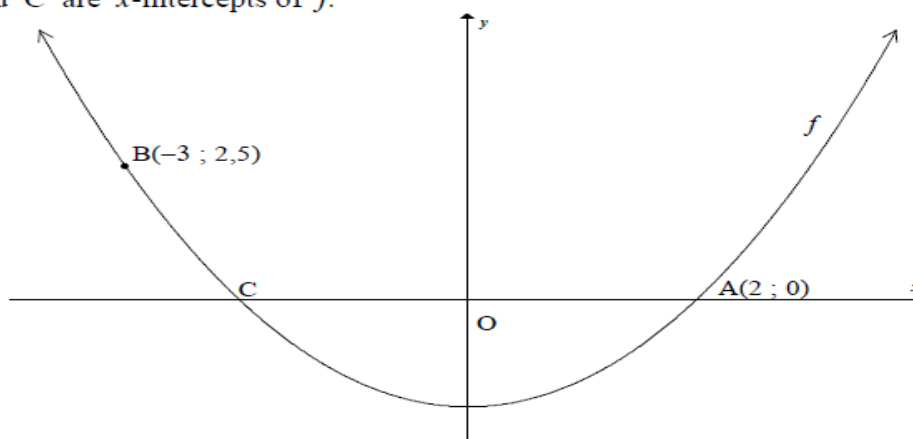


Solve the following

The graph of $f(x) = ax^2 + q$ is sketched below.

Points A(2 ; 0) and B(-3 ; 2,5) lie on the graph of f .

Points A and C are x-intercepts of f .



Write down the coordinates of C.

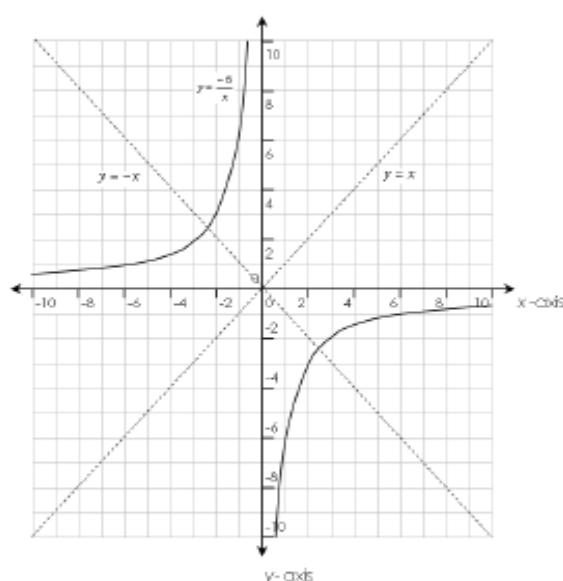
Determine the equation of f .

Write down the range of f .

HYPERBOLA GRAPH

1. $y = -\frac{6}{x}$

x	-6	-4	-3	-2	-1	0	1	2	3	4	6
y	1	$\frac{3}{2}$	2	3	6	0	-6	-3	-2	$-\frac{3}{2}$	-1



- The worked example's graph occupies the 1st and 3rd quadrants, while this (otherwise identical) graph occupies the 2nd and 4th quadrants. The graph is reflected about the x-axis. The negative sign has made all the difference.
- See above. The hyperbolic function is bisected by the line $y = -x$. The line cuts the graph into two identical halves.

The exponential function

The exponential graph requires a lot of practice. Ensure that the students know the procedure for plotting such a graph. They need to know the general form of the graph.

Both $y = 2^x$ and $y = 3^x$ have the same y-intercept: $y = 1$. Furthermore, they both approach – yet do not cross or touch – the line $y = 0$. Neither graph has an x-intercept. The slope of the graph of $y = 3^x$ increases at a greater rate than $y = 2^x$.

